■ GAM Assurance Hackathon

Complete Student Guide — What To Do & How

The Big Picture: GAM Assurance insures thousands of properties (homes, shops, factories) across Algeria. They are worried about earthquakes — Algeria is one of the most seismically active countries in the world. Right now, GAM has no clear picture of which areas carry the most risk. **Your job: give them that picture, and tell them what to do about it.**

1. What Is the Problem? (Plain Language)

GAM Assurance collects insurance premiums from clients all over Algeria. If an earthquake hits a city where GAM has many insured properties, GAM could have to pay out enormous sums of money all at once — maybe more money than they can afford. This is called **portfolio vulnerability**.

Three specific problems they identified:

- **No seismic map overlay:** They don't know which of their insured properties are in dangerous earthquake zones.
- **No concentration analysis:** They might have too many clients in one risky city and too few in a safe city — bad for balance.
- **No financial impact estimate:** If a big earthquake happened today, how much would GAM have to pay? They have no answer right now.

2. What Data Are You Given?

File / Source	What It Contains	How You Use It
CATNAT_2023_2025.xlsx	GAM's actual insurance portfolio: commune codes, insured amounts, number of policies, claims history	This is the CORE DATA. Every analysis starts here.
code_geo_fra_ara.pdf	National geographic codes: Wilaya number → Wilaya name, Commune code → Commune name	Use to link commune codes in the Excel to real place names & map coordinates.
Algerian_RPA99.pdf	Algeria's official seismic building code. Contains the seismic zone map & zone classification by Wilaya/Commune.	Extract seismic zone (0, I, IIa, IIb, III) for every commune. Zone III = most dangerous.

File / Source	What It Contains	How You Use It
Cahier_de_charge.docx	Official event brief / specifications from the organizers	Read carefully — contains scoring criteria & deliverable requirements.

3. What Exactly Must You Deliver?

The organizers described **4 deliverables**. Here is each one explained simply:

Deliverable A — Interactive Risk Map

What it is: A map of Algeria (interactive, like Google Maps) where each commune is colored by its seismic danger level:

- ■ Green = Zone 0 / I (low risk)
- ■ Orange = Zone IIa / IIb (medium risk)
- ■ Red = Zone III (high risk)

Then you OVERLAY GAM's portfolio on top of this map. Each commune shows: how many policies GAM has there, and the total insured amount.

What the map reveals:
— *Over-concentration zones:* communes where GAM has too many clients AND the seismic risk is HIGH (dangerous).
— *Under-concentration zones:* communes where GAM has very few clients but risk is LOW (missed business opportunity).

Deliverable B — AI Risk Classification (CatBoost)

What it is: A machine-learning model that automatically classifies every commune into a risk score: ■ Red / ■ Orange / ■ Green.

Why CatBoost specifically? The organizer said Random Forest won't work here because the problem involves categorical data (zone names, commune types). CatBoost is designed for categorical features — it handles them natively without encoding tricks.

Input features for the model (examples):
• Seismic zone from RPA99 (Zone 0, I, IIa, IIb, III)
• Number of insured properties per commune
• Total insured capital (sum assured)
• Historical claims frequency (from CATNAT data)
• Property type (residential / commercial / industrial)

Output: A score label (Red / Orange / Green) for each commune, with a probability/confidence value.

Deliverable C — Monte Carlo Catastrophe Simulation

What it is: A computer simulation that asks: *"If a major earthquake happened, how much would it cost GAM?"*

Key rule from the organizer: The simulated earthquake CANNOT happen in a Green zone. It must be placed only in Orange or Red zones — because that's where real earthquakes are most likely.

How Monte Carlo works (simple version):
1. Run the simulation 10,000+ times
2. Each time, randomly pick an Orange or Red commune (weighted by danger level)
3. Randomly pick an earthquake magnitude
4. Calculate the damage to all GAM properties in the affected area using a damage function
5. Record the total payout needed
6. After all runs, show: average loss, worst-case loss (99th percentile = PML), distribution chart

The key output number is called PML (Probable Maximum Loss) — the maximum amount GAM would realistically need to pay in a catastrophic event.

Deliverable D — Reinsurance Strategy Recommendations

Context: GAM already cedes 70% of its CatNat (natural catastrophe) revenue to reinsurers. This means: for every 100 DZD of premium collected, GAM keeps 30 DZD and transfers 70 DZD to a reinsurance company, which takes on 70% of the risk.

What you must do: Based on your map and Monte Carlo results, propose concrete reinsurance strategies to better protect the remaining 30%.

Strategies to consider (you don't need all, pick the best-fit ones):

- **Excess of Loss (XL):** Reinsurer pays when a single event exceeds a threshold (e.g., above 500M DZD)
- **Catastrophe Aggregate Cover:** Covers accumulated losses from multiple events in one year
- **Quota Share adjustment:** Increase the % ceded in high-risk regions, reduce in low-risk
- **Geographic diversification:** Recommend growing the portfolio in Green zones to naturally reduce risk

Support each recommendation with your data (e.g., 'Zone III communes represent X% of portfolio but carry Y% of expected losses').

4. Step-by-Step: What To Do (Your Action Plan)

1	**Read All Documents First** Read the Cahier de charge (brief) carefully. Understand the scoring criteria. Skim the RPA99 document to find the seismic zone table (Annexe 1 at the end — it lists every Wilaya and commune with its zone number). Understand the Excel data structure.
2	**Build the Seismic Zone Database** From the RPA99 Annexe 1, extract the seismic zone for all 48 Wilayas and their communes. Create a table: [Wilaya_code | Commune_code | Wilaya_name | Commune_name | Seismic_zone]. Use the code_geo PDF to get the commune codes. This step is tedious but critical.
3	**Load & Clean the Excel Portfolio Data** Open CATNAT_2023_2025.xlsx in Python (pandas). Check columns: commune code, policy count, insured capital, claim amounts. Merge your seismic zone table onto this data using the commune code as the join key. Result: one dataframe with both portfolio and seismic risk info per commune.
4	**Build the Interactive Map (Folium or Plotly)** Use Python's folium or plotly library with Algeria's GeoJSON boundaries. Color each commune by seismic zone (Red/Orange/Green). Add markers or choropleth shading for portfolio concentration (size = insured capital). Add popups showing details per commune. Export as an HTML file. This is your Deliverable A.
5	**Train the CatBoost Classifier** Features: seismic zone, nb_policies, sum_insured, claim_ratio, region_type, etc. Label: risk class (you can define labels from seismic zone + concentration). Split data (train/test), train CatBoost, evaluate with confusion matrix. Use the model to assign Red/Orange/Green scores. This is Deliverable B.
6	**Run the Monte Carlo Simulation** For each simulation run (10,000+): (a) Sample an epicenter from Orange/Red communes (weighted by zone severity). (b) Sample a magnitude from a realistic distribution. (c) Apply a loss function to estimate % damage per property type. (d) Multiply by insured capital. (e) Record total loss. Plot the loss distribution. Report PML (99th percentile). This is Deliverable C.
7	**Write the Reinsurance Recommendations** Compare your PML against GAM's current 30% retained risk exposure. Identify which communes/regions are most dangerous. Propose specific treaty structures. Back everything with numbers from your simulation. Keep it practical — what can GAM actually do next year? This is Deliverable D.
8	**Build the Final Dashboard** Combine everything into one interactive dashboard (Dash, Streamlit, or Power BI). It should show: the map, the AI scores, key KPIs (total portfolio at risk, PML, over/under-concentration zones), and a recommendations panel.

5. Key Technical Concepts Explained Simply

What is Seismic Zone III?

Algeria's RPA99 code divides the country into 5 seismic zones (0, I, IIa, IIb, III). Zone III is the most dangerous — buildings there face the strongest potential earthquakes. For example, Blida and Boumerdes (site of the 2003 earthquake) are Zone III. Zone 0 includes the deep Sahara — very low risk.

What is Portfolio Concentration?

If 40% of GAM's insured capital is in Algiers and an earthquake hits Algiers, GAM loses 40% of its portfolio in one event. That's over-concentration. A well-diversified insurer spreads risk — not too much in any one zone.

What is PML (Probable Maximum Loss)?

PML is the maximum loss an insurer would expect in a realistic worst-case scenario. It's usually defined as the 99th percentile of the Monte Carlo loss distribution — meaning: 'In 99% of simulated catastrophes, the loss stays below this number.' Insurance regulators require companies to hold enough capital to cover their PML.

Why CatBoost and not Random Forest?

Random Forest converts categorical variables (like zone names) into numbers via one-hot encoding, which can be inefficient with many categories. CatBoost handles categorical features directly using target statistics — it's faster, more accurate on this type of data, and needs less preprocessing.

6. Recommended Tools & Libraries

Task	Recommended Tool	Why?
Data wrangling	Python + pandas	Fast, powerful, widely known
Interactive map	Folium or Plotly+Dash	Easy to build, HTML output, interactive
AI classification	CatBoost (pip install catboost)	Required by organizers for categorical data
Monte Carlo simulation	NumPy + SciPy	Standard scientific computing in Python
Dashboard	Streamlit or Dash	Build a web app in Python, no HTML needed
Seismic zone data	RPA99 Annexe 1 (PDF)	Official Algeria government source
Map boundaries	Algeria GeoJSON (open data)	Search 'Algeria communes geojson GitHub'

7. What the Final Dashboard Should Show

The organizer's final objective is a **dashboard** that a non-technical insurance manager can open and immediately understand the situation. It should contain:

Panel 1 — Interactive Map: Algeria colored by seismic risk + GAM portfolio overlay
Panel 2 — Concentration Analysis: List of over-concentrated zones (Red flags) and under-concentrated zones (opportunities)
Panel 3 — AI Risk Scores: Table of communes with Red/Orange/Green labels from CatBoost
Panel 4 — Monte Carlo Results: Loss distribution chart, PML value, expected annual loss
Panel 5 — Recommendations: Reinsurance strategy suggestions with supporting data

8. Quick Summary: What You Are Building

Deliverable	In Simple Words	Key Output
A – Risk Map	A map showing where GAM's money is vs where earthquakes are likely	Interactive HTML map (Folium/Plotly)
B – AI Scoring	A machine-learning model that scores every commune Red/Orange/Green	CatBoost model + scored table
C – Simulation	A computer experiment: 'What if a big earthquake hits today? How much does GAM lose?'	PML value + loss distribution chart
D – Strategy	Based on C: recommendations to protect GAM's portfolio	Written report + treaty proposals
Final – Dashboard	One screen that shows all of the above in a clear, visual way	Streamlit/Dash web app

■ **Pro tip:** Start with Step 2 (building the seismic zone database) since everything else depends on it. The RPA99 Annexe 1 already lists every Wilaya and commune — just extract it systematically. The commune geographic codes PDF is your key to linking the Excel data to real place names and map coordinates. Good luck!